STAT184 Final Project

Abhiraj Srivastava

Table of contents

1 Introduction

This project explores trends in public confidence toward medicine using data from the General Social Survey (GSS). The analysis focuses on how trust in medical institutions changed between 2018 and 2024, particularly before and after the COVID-19 pandemic. Public trust in medicine is an important social indicator because it affects healthcare behavior, vaccine uptake, and confidence in scientific institutions.

2 Data Provenance

The dataset used in this project comes from the General Social Survey (GSS), a nationally representative survey conducted in the United States. The GSS collects demographic, behavioral, and attitudinal data from respondents across the country.

The specific variable used in this analysis is:

- `CONMEDIC` — confidence in medicine

Additional variables from the GSS may also be used for exploratory analysis and comparison.

3 FAIR and CARE Principles

3.1 FAIR Principles

The dataset partially satisfies the FAIR principles:

- **Findable:** The GSS dataset is publicly available and well documented.
- **Accessible:** Researchers and students can access the data online.
- **Interoperable:** The dataset works with statistical software such as R.
- **Reusable:** Documentation and variable descriptions allow future reuse.

3.2 CARE Principles

The CARE principles are only partially satisfied:

- The GSS provides broad demographic representation.
- However, the data are not specifically designed around Indigenous data governance or community ownership principles.

4 Data Cleaning and Wrangling

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.0     v readr     2.1.6
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```r
library(ggplot2)

GSS_clean <- readRDS("GSS_clean.rds")
```

The data required cleaning because missing values in the GSS are often represented with negative numbers such as `-100`. These values were converted into `NA` values before analysis.

```r
medical_data <- GSS_clean |>
  filter(YEAR %in% c(2018, 2021, 2022, 2024)) |>
  mutate(
    CONMEDIC = if_else(CONMEDIC < 0, NA_integer_, CONMEDIC)
  )
```

5 Exploratory Data Analysis

The exploratory analysis focuses on trends in medical confidence across multiple years.

5.1 Medical Confidence Table

```r
medical_table <- medical_data |>
  mutate(
    Medical_Confidence = case_when(
      CONMEDIC == 1 ~ "High Trust / Full Confidence",
      CONMEDIC == 2 ~ "Moderate / Skeptical",
      CONMEDIC == 3 ~ "Low / No Confidence",
      TRUE ~ NA_character_
    )
  ) |>
```

```
  drop_na(Medical_Confidence) |>
  count(YEAR, Medical_Confidence) |>
  group_by(YEAR) |>
  mutate(
    Percent = round(n / sum(n) * 100, 1)
  ) |>
  ungroup()

knitr::kable(
  medical_table,
  caption = "Medical Confidence by Year",
  align = "c"
)
```

Table 1: Medical Confidence by Year

YEAR	Medical_Confidence	n	Percent
2018	High Trust / Full Confidence	553	35.6
2018	Low / No Confidence	211	13.6
2018	Moderate / Skeptical	790	50.8
2021	High Trust / Full Confidence	1070	40.2
2021	Low / No Confidence	246	9.2
2021	Moderate / Skeptical	1346	50.6
2022	High Trust / Full Confidence	786	33.4
2022	Low / No Confidence	373	15.8
2022	Moderate / Skeptical	1196	50.8
2024	High Trust / Full Confidence	569	26.4
2024	Low / No Confidence	428	19.8
2024	Moderate / Skeptical	1162	53.8

6 Data Visualization

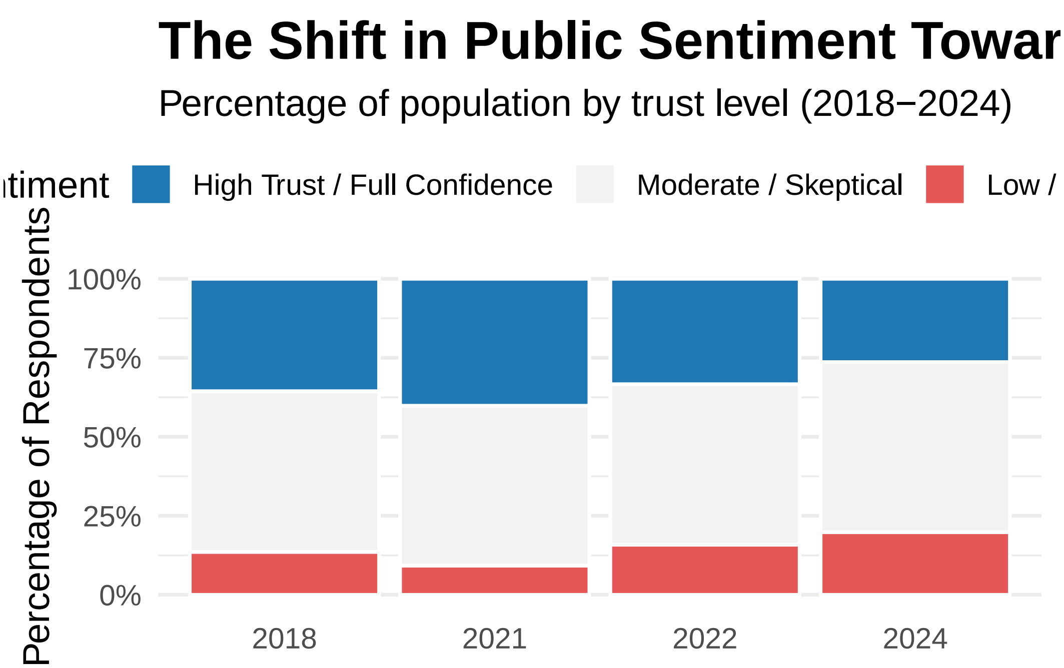
6.1 Stacked Bar Chart of Medical Confidence

```
medical_graph_data <- medical_data |>
  mutate(
    Sentiment = case_when(
      CONMEDIC == 1 ~ "High Trust / Full Confidence",
```

```
      CONMEDIC == 2 ~ "Moderate / Skeptical",
      CONMEDIC == 3 ~ "Low / No Confidence",
      TRUE ~ NA_character_
    )
  ) |>
  drop_na(Sentiment) |>
  count(YEAR, Sentiment) |>
  group_by(YEAR) |>
  mutate(percent = n / sum(n)) |>
  ungroup() |>
  mutate(
    Sentiment = factor(
      Sentiment,
      levels = c(
        "High Trust / Full Confidence",
        "Moderate / Skeptical",
        "Low / No Confidence"
      )
    )
  )

ggplot(
  medical_graph_data,
  aes(
    x = factor(YEAR),
    y = percent,
    fill = Sentiment
  )
) +
  geom_col(
    width = 0.9,
    color = "white"
  ) +
  scale_y_continuous(
    labels = function(x) paste0(x * 100, "%"),
    limits = c(0, 1)
  ) +
  scale_fill_manual(
    values = c(
      "Low / No Confidence" = "#e45756",
      "Moderate / Skeptical" = "#f2f2f2",
      "High Trust / Full Confidence" = "#1f77b4"
```

```
    )
  ) +
  labs(
    title = "The Shift in Public Sentiment Toward Medicine",
    subtitle = "Percentage of population by trust level (2018-2024)",
    x = NULL,
    y = "Percentage of Respondents",
    fill = "Sentiment"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(
      face = "bold",
      size = 20
    ),
    plot.subtitle = element_text(size = 14),
    legend.position = "top",
    panel.grid.major.x = element_blank(),
    panel.grid.minor.x = element_blank()
  )
```



“‘{r}
R code goes here

summary(cars)
“‘

7 Interpretation of Results

The visualization suggests that public confidence in medicine shifted noticeably after 2020. The percentage of respondents expressing low confidence increased over time, while high trust gradually declined. This trend may reflect the political and social polarization surrounding public health during and after the COVID-19 pandemic.

8 Conclusion

This analysis demonstrated how confidence in medicine changed between 2018 and 2024 using data from the General Social Survey. Through data cleaning, exploratory analysis, tables, and visualizations, the report highlighted changing public sentiment toward medical institutions in the United States.

9 Code Appendix

```r
sessionInfo()
```

```
R version 4.5.2 (2025-10-31)
Platform: aarch64-apple-darwin20
Running under: macOS Ventura 13.5

Matrix products: default
BLAS:   /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framewo
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;

locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: America/New_York
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base
```

```
other attached packages:
 [1] lubridate_1.9.5 forcats_1.0.1   stringr_1.6.0   dplyr_1.2.0
 [5] purrr_1.2.1     readr_2.1.6     tidyr_1.3.2     tibble_3.3.1
 [9] ggplot2_4.0.2   tidyverse_2.0.0

loaded via a namespace (and not attached):
 [1] gtable_0.3.6       jsonlite_2.0.0     compiler_4.5.2     tidyselect_1.2.1
 [5] scales_1.4.0       yaml_2.3.12        fastmap_1.2.0      R6_2.6.1
 [9] labeling_0.4.3     generics_0.1.4     knitr_1.51         pillar_1.11.1
[13] RColorBrewer_1.1-3 tzdb_0.5.0         rlang_1.1.7        stringi_1.8.7
[17] xfun_0.56          S7_0.2.1           otel_0.2.0         timechange_0.4.0
[21] cli_3.6.5          withr_3.0.2        magrittr_2.0.4     digest_0.6.39
[25] grid_4.5.2         rstudioapi_0.18.0  hms_1.1.4          lifecycle_1.0.5
[29] vctrs_0.7.1        evaluate_1.0.5     glue_1.8.0         farver_2.1.2
[33] rmarkdown_2.30     tools_4.5.2        pkgconfig_2.0.3    htmltools_0.5.9
```

{r}